



PEPSICO: DALLAS GATORADE PLANT WATER REDUCTION UPGRADES

SECTOR

Industrial

LOCATION

Dallas

BACKGROUND

Gatorade is a business unit of PepsiCo Beverages North America, and its Dallas Gatorade plant completed two major water efficiency projects to meet PepsiCo's sustainability and water efficiency goals. PepsiCo has designated the Dallas site as falling within a designated high-water-risk (HWR) area, which is defined by the company's global water risk assessment process using the World Resources Institute's Aqueduct tool.



2022 BETTER PROJECT WINNER

For all HWR manufacturing sites, PepsiCo set a goal to achieve water use efficiency equal to 1.2 Gallons of water per gallon of production in beverage plants. For PepsiCo facilities, this equates to more than a 50% reduction in absolute water use from the 2015 baseline year. Additionally, the Dallas Gatorade site has set a goal to replenish more than 100% of the water that they use back into the local watershed by 2030.

SOLUTIONS

PepsiCo developed two main projects—one focused on water filtration system improvements and the other focused on reverse osmosis (RO) concentrate recovery. The project is expected to reduce incoming water usage by approximately 30 million gallons, based on 2017 as the baseline year. Due to the complexity and scope of the project, it was executed in several phases. It was completed in the summer of 2021.

The initial phase of the project involved recovering reverse osmosis (RO) concentrate from the site's product water filtration system. The recovered concentrate was softened to remove calcium and magnesium and was then used to supply makeup water to the site's cooling towers. The cooling towers provide process cooling for all of the manufacturing lines at the Dallas Gatorade site, as well as air compressor cooling and HVAC heat rejection. To use softened recovered water in the cooling towers, the cooling tower water treatment program was changed from using chemical-based city water as makeup water (MUW) to a soft water MUW system. The overall time frame for this initial phase of the project took 4 years to complete and confirm performance (after several years of service all the heat exchangers were disassembled and inspected for scale formation).

The other phase of the project focused on measures to improve the efficiency of the site's water filtration system. This phase involved recovering backwash from the site's pre-filters and carbon filters, then re-filtering it and sending it back to the water room for re-processing. The incoming water to the water room was processed with a water softener to remove hardness, which allowed the reverse osmosis efficiency to increase from 80% to 95%.

OTHER BENEFITS

The success of the Dallas site's project spurred a culture of water reduction that led to the identification of numerous other water efficiency projects at the site and will be replicated across all of PepsiCo Beverages North America's HWR sites over the next five years. The Dallas facilities and project management team worked closely with the maintenance and operations team to implement new water efficiency best practices and developed a cellphone application that enables the team to communicate water efficiency opportunities and necessary repairs.

With the softening of the RO feedwater, RO concentrate now has less hardness, which takes some of the hardness removal load off the recovery system softener. Additionally, the implementation of measures on the cooling towers has heightened awareness of the cooling tower operation, which has dramatically improved the reliability of cooling tower water chemistry and operation. Since this project was completed, there has also been a reduction in the downtime associated with process cooling, which has helped improve the site's productivity.